

Parallel Morphological Image Processing

Accelerating morphological operations via C++17 and OpenMP

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Da un secolo, oltre.

Introduction



- **Mathematical Morphology:** framework for geometric feature extraction and image restoration
- **Targeted Operations:**
 - *Fundamental:* Erosion, Dilation
 - *Compound:* Opening, Closing
- **Goal:** resolve computational bottlenecks in high-resolution image processing
- **Technology Stack:** C++17 and OpenMP for thread orchestration
- **Dataset & Benchmarks:** BSDS500
 - Preliminary grayscale conversion
 - 2.0x and 4.0x upscaling to evaluate scalability

Context & Architectural Limits

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Operator Logic

- **Erosion:** minimum extraction in a defined spatial neighborhood
- **Dilation:** maximum extraction in a defined spatial neighborhood
- **Opening:** Erosion followed by Dilation
- **Closing:** Dilation followed by Erosion

The Memory Wall & Complexity

- Naive execution with 2D stencils strictly bound by $\mathcal{O}(M \times N \times K^2)$ complexity
- **Compound operations** implemented sequentially (double-pass approach)
- **System Overhead:** destruction of *temporal locality* and cache saturation between passes

Hardware Optimization: Pixel-Level Parallelism



- Spatial stencil processing is inherently independent for each pixel
- **Loop Collapsing** (`collapse(2)`): logical fusion of X and Y image axes
 - Expands the iteration space and distributes workload with fine granularity
 - Prevents *thread starvation* at boundaries
- **Memory Safety** (`default(none)`): forced explicit scoping
 - Thread-local allocation for temporary variables (e.g., `min_val`)
 - Avoids synchronization overhead (zero locks or atomics) in the critical loop
- **Static Scheduling** (`schedule(static)`): deterministic work distribution
 - Zero runtime overhead, ideal for uniform per-pixel workloads (invariant $K \times K$ footprint)
 - Note: dynamic chunking later explored

Algorithmic Optimization: 1D Separable Kernels



- **The Dominant Lever:** decomposition of the $K \times K$ stencil into two independent passes (Horizontal $\rightarrow$ Vertical)
- **Computational Reduction:** transition from $\mathcal{O}(K^2)$ to linear $\mathcal{O}(2K)$
 - Using a 15×15 kernel, base operations per pixel drop from 225 to just 30

The Architectural Trade-off (Cache Asymmetry)

- **Horizontal Pass:** highly efficient, exploits prefetching and RAM *spatial locality* (row-major layout)
- **Vertical Pass:** introduces severe cache misses, forcing the memory bus to work column-wise

Producer-Consumer Pipeline



- **The Fork-Join Problem:** pure 1D Closing requires 4 global passes, doubling heap footprint and introducing strict synchronization barriers
- **OpenMP Pipeline Design:**
 - Single partitioned parallel region where half the threads act as Producers and half as Consumers
 - Fine-grained chunk synchronization via PaddedFlag array to eliminate false sharing
- **Architectural Benefits:** execution overlap and keeps pixels in cache (*temporal locality*)

Performance Results

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Strong & Weak Scalability Analysis

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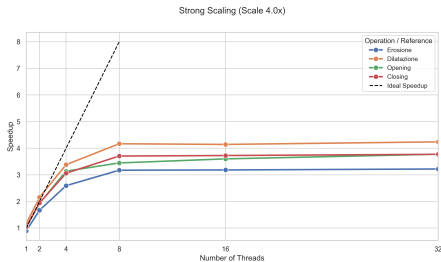


Figure: Strong Scaling (Fixed Workload)

- Asymptotic saturation at **3.5x-4.0x** with 4 physical cores
- Perfectly reflects *Amdahl's Law* (unavoidable pure sequential fraction)

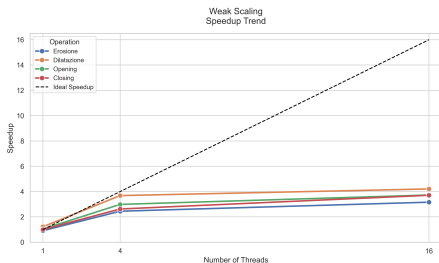


Figure: Weak Scaling (Scaled Workload)

- Efficiency drops from **1.0** (1 thread) to **0.73** (4 threads)
- Collapses to **0.23** at 16 threads: hyper-threading and RAM bandwidth limits

Algorithmic Reduction: 1D vs 2D Performance

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- **Direct Impact:** transitioning to linear $\mathcal{O}(2K)$ complexity acts as the true performance multiplier
- **Kernel Scaling (K):**
 - At low K values, the gap is minimal
 - At high K values, the 2D approach collapses under quadratic load

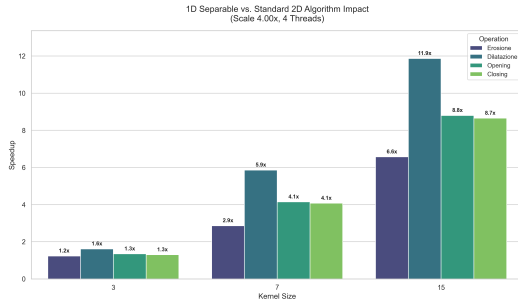


Figure: Speedup of 1D Separable Kernels vs 2D baseline

Producer-Consumer Pipeline Efficiency

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- **Focus:** specific optimization for Opening and Closing
- **Peak Efficiency:** yields up to a **1.52x** performance boost over the standard parallel baseline
- **High-Concurrency Contention:**
 - Active *spin-wait* loops saturate physical cores under high logical thread counts
 - Cross-chunk dependencies amplify synchronization overhead and cause thread starvation

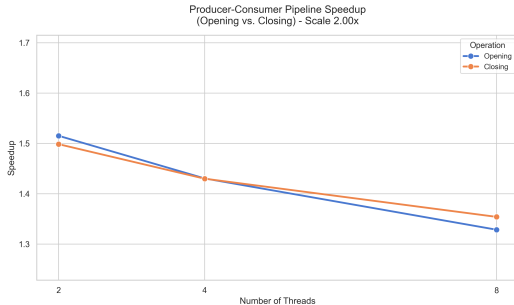


Figure: Pipeline speedup for compound operations

Other Strategies Tested



- **SIMD Vectorization** (`#pragma omp simd`):
 - No positive impact on core loops
 - Failure due to compiler auto-vectorization (already optimal) and min/max conditional branches
- **Dynamic Loop Scheduling** (`schedule(dynamic)`):
 - Evaluated on the baseline parallel version to improve load balancing
 - Performance remained virtually identical to the default `static` approach

Peak Performance & Future Developments

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Optimal Configuration 22x Peak Speedup

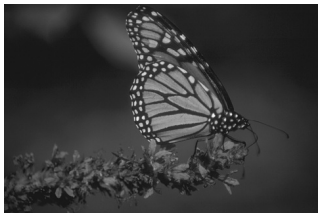
Mixed 1D Separable + OpenMP
architecture ($K = 15$, 4 Threads) over
sequential version

- **Hardware/Software Conclusion:** bare multithreading can't save a flawed algorithm. Hardware/software integration is the only path to raise high performance
- **Heterogeneous Computing (Future Work):**
 - Porting the logic to GPU architectures (CUDA)

Visual Results

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Original



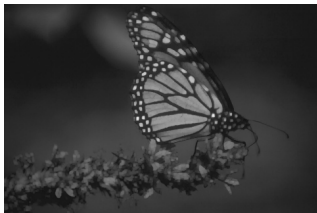
Erosion



Dilation



Opening



Closing



Parallel Morphological Image Processing

Thank you for listening!
Any questions?